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Rescue and safety system development and performance evaluation by network reliability

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ABSTRACT

During the disaster, a rescue and safety system (RSS) must be developed to evaluate whether distributing rescue time or increasing the number of ambulances can successfully transport casualties. The rescue routes for ambulance transport can be represented as a network with arcs denoting the rescue roads and nodes denoting the location of casualties and casualty collection points (CCP). The uncertain damage on rescue roads causes the RSS hard to evaluate the rescue efficiency. Therefore, the transport time of ambulance is considered stochastic and thus such rescue network can be modeled as a stochastic rescue network (SRN). Network reliability of an SRN is defined as the probability that the number of casualties can be successfully transported from the affected area to CCP in a rescue time. An SRN model is constructed according to the road information under earthquake, and then a time probability table is established by the rescue road vehicle density to represent the traffic. A practical case in Taiwan is demonstrated to show the applicability of proposed network reliability evaluation. According to the information obtained from the proposed algorithm, the RSS can be further developed into a smartphone application to provide the commander with several suggestions of rescue operations.

1. Introduction

Several damage has been caused in Taiwan by the earthquakes in the past, such as the Hsinchu-Taichung earthquake in 1935 and the 921 earthquakes in 1999 [1,2]. According to the research conducted by the Taiwan Earthquake Research Center [3], the probability of a 6.5 magnitude earthquake occurring within 50 years is greater than 50 percent, while the difficulty of rescue increases with the subsequent influence of the earthquake. For instance, the number of casualties increases due to secondary disaster such as collapsed building, fire and even flooding. It becomes a challenge for establishing the rescue and safety system (RSS) to effectively be aware of the damage in the rescue operation. As the earthquake occurs, an RSS must provide the decision suggestions of dispatching the ambulance to transport the casualties. Therefore, the way to evaluate the rescue efficiency of the rescue operation is a concern to determine whether the number of ambulances or rescue time is enough. In other words, developing a performance index to measure rescue efficiency is an important issue for developing an RSS.

In a rescue operation [4–6], the rescue network is adopted and each rescue road is defined as an arc. The optimization problem of the rescue network has been widely studied, such as distribution planning [7,8] and location allocation [9–11]. Furthermore, the rescue routing problem is a critical issue that has been implemented by different optimization model. By using the objective function to maximize rescue efficiency, Liu et al. [12] applied the integer programming model to determine the optimal rescue route. Shahparvari and Abbasi [13] proposed a greedy algorithm to determine the routes required in the evacuee population. For the minimization function, Lim et al. [14] adopted the mixed integer programming method to minimize the jam density of the rescue route in the evacuation route allocation problem. Wang et al. [15] combined the Lagrangian relaxation-based approach with a heuristic algorithm to find the prior evacuation plans by minimizing the expected total evacuation time. Dulebenets et al. [16] constructed the mixed-integer programming model to determine the evacuation routes to assign the casualties by minimizing the total travel time of casualties leaving an evacuation zone. In summary, the route allocation problem can be solved by optimization models. However, the damage of the rescue road would cause the transport time of the ambulance to be stochastic and need to be

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Acronyms

RSS	rescue and safety system
SRN	stochastic rescue network
MP	minimal path
CCP	casualty collection point
RSDP$^+$	modified recursive sum of disjoint product
D-MDV	maximal duration vector satisfying demand D
TERIA	Taiwan earthquake impact research and information application platform

Notations

n	number of rescue roads (arcs)
m	number of MP
c	number of ambulances
v	number of pseudo maximal duration vector
a_i	rescue road i (arc), $i = 1, 2, \ldots, n$
$\mathbf{A}$	$\{a_1, a_2, \ldots, a_n\}$: set of arcs
w_i	road width of a_i, $i = 1, 2, \ldots, n$
$\mathbf{W}$	$\{w_1, w_2, \ldots, w_n\}$: set of road width
l_i	road length of a_i, $i = 1, 2, \ldots, n$
$\mathbf{L}$	$\{l_1, l_2, \ldots, l_n\}$: set of road length
p_i	probability of road failure of a_i, $i = 1, 2, \ldots, n$
$\mathbf{P}$	$\{p_1, p_2, \ldots, p_n\}$: set of road failure
D	demand of casualties
v_i	average ambulance speed of a_i, $i = 1, 2, \ldots, n$
v_f	ambulance speed without influences from other vehicles.
d	average vehicle density of road
d_i	jam density of a_i, $i = 1, 2, \ldots, n$
φ	density modified parameter used in Pipes' generalized model
t_i	transport time of a_i, $i = 1, 2, \ldots, n$
t_i^o	transport time of a_i in undamaged state, $i = 1, 2, \ldots, n$
t_i^m	transport time of a_i in moderate damage state, $i = 1, 2, \ldots, n$
t_i^s	transport time of a_i in severe damage state, $i = 1, 2, \ldots, n$
p_i^o	probability of road failure of a_i in undamaged damage state, $i = 1, 2, \ldots, n$
p_i^m	probability of road failure of a_i in moderate damage state, $i = 1, 2, \ldots, n$
p_i^s	probability of road failure of a_i in severe damage state, $i = 1, 2, \ldots, n$
θ	average transport time of each trip
z_i	pseudo maximal duration of a_i, $i = 1, 2, \ldots, n$
Z	$(z_1, z_2, \ldots, z_n)$: pseudo maximal duration vector
x_i	current duration of a_i, $i = 1, 2, \ldots, n$
X	$(x_1, x_2, \ldots, x_n)$: current duration vector
T	transport time constraint
η	number of trips for ambulance transporting the casualties
$\mathbf{X_D}$	set of duration vectors satisfying demand D
R_D	network reliability for demand D
R_j	jth MP in the SRN, $j = 1, 2, \ldots, m$
E_R	expected transport time of rescue route R
$\mathbf{\Omega}$	set of D-MDV candidates
$\mathbf{\Omega_{max}}$	set of maximal ones from $\mathbf{\Omega}$

further considered in the rescue route problem. This study constructs the rescue operation into a stochastic rescue network (SRN) for ambulance transport by considering the stochastic transport time of the ambulance, while the rescue operation, in which casualties are transported to the casualty collection point (CCP) is regarded in this study. In such network, each arc is denoted as a rescue road where the source node is the location of casualties and the sink node is the location of the CCP. Several systematic approaches have been used to evaluate network reliability based on two key methods: minimal path (MP) [17–20] and minimal cut (MC) [19,21-23]. An MP is defined as a set of arcs from the source to the sink node with no cycle. An MC is a cut that will not be a cut when removing any arc from it. In order to ensure the shortest transport time for the ambulance, the concept of MP [17,18] is employed as the rescue route of the ambulance.

While planning a rescue operation, reliability is an important performance indicator utilized to assess rescue efficiency. Various studies have explored reliability issues in emergency routing problems. For instance, Wang et al. [24] used a nominated sorting genetic algorithm to maximize the total reliability of the route connection. Wang et al. [25] constructed the reliability prediction model to evaluate the evacuation route on the offshore platform. Several papers also studied the reliability on the transportation system [26–28]. Kim and Song [26] proposed an integrated indicator that considers the accessibility and reliability to evaluate the resilience and vulnerability of the spatial network. Zhang et al. [27] built the railway container chain evaluation model to calculate the probability that goods can be transported in specified distribution within expected travel time, which is reliability. However, the uncertainty of the rescue road has not been taken into consideration in previous studies' reliability definition. At the same time, the number of casualties transported to the CCP in a given time cannot be obtained from previous studies. To address the mentioned problem, the stochastic transport time is studied in an SRN and the network reliability is further developed as the performance index to evaluate the rescue efficiency. Network reliability is defined as the probability that the number of casualties can be successfully transported from the affected area to the CCP in a given rescue time by ambulance. Such an indicator can help the development of RSS to make more informed decisions during rescue operations. The rescue efficiency within a rescue operation can be measured by quantifying the network reliability with different factors. For example, if an allocation is used less rescue time while maintaining the same network reliability, and we would say that is more efficient. In other words, the rescue efficiency is quantified by the proposed network reliability in this study to develop an RSS.

The concept of network reliability is applied in many fields, such as supply chain [20,29,30], manufacturing system [31,32], computer network [33–35], and project management [36–38]. In the field of transportation systems [39,40], Lin and Nguyen [41] calculated the network reliability for the stochastic flight network considering the arrival time and the number of stopovers. Lin et al. [42] evaluated the air transport network reliability with various source and sink stations. Yeh et al. [43] constructed the stochastic rail transport network to calculate the probability that the number of passengers can be successfully transported to the station. These studies modeled real transportation system as a stochastic network and measured the network reliability in terms of MP as a performance index. To our best knowledge, evaluation of the network reliability of SRN involving the stochastic transport time under earthquake situation is firstly studied in this paper. However, the stochastic factor under damage for each road needs to be further studied to better present disaster situations in real-world network. In summary, the damage condition of each road directly impacts the efficiency of ambulance transport. To address such a problem, an SRN is established using network analysis method, which incorporates the stochastic transport time associated with each road. By evaluating the network reliability of the SRN based on MP method, the efficiency of a rescue operation can be assessed. The ambulance route and rescue time in a rescue operation can be suggested based on the network reliability. Incorporating network reliability into the development of an RSS serves the main purpose of enhancing the safety of transport process and offering practical solutions for rescue operations.

The remainder of this paper is organized as follows. The construction of stochastic rescue network and network reliability calculation are described in Section 2. Section 3 introduces an algorithm to calculate the network reliability. A practical case in Tainan, Taiwan is adopted to demonstrate the applicability in Section 4. Finally, the discussion and conclusion are summarized in Section 5.

2. Development of a stochastic rescue network model

When an earthquake occurs, the commander needs to acquire information from the available route linking a source effected area and a sink CCP in accordance with the damage to the road and then select the appropriate route for the ambulance to rescue the casualties. Using the network analysis approach, data is collected from the Taiwan Earthquake impact Research and Information Application platform (TERIA) and the effective data is filtered and processed to construct an SRN for different disaster situation. The network reliability of an SRN is adopted to measure whether the injured group can be successfully delivered to the CCP through available ambulances. In Section 2.1, the rescue network is introduced to describe the rescue operation. The establishment of the time probability table is presented in Section 2.2. Finally, the calculation of network reliability is proposed in Section 2.3.

2.1. Construction of a rescue network for casualties

In emergency, the selection of the rescue road is quietly important to ensure the accessibility of the road and can be further transformed to a rescue network (**A, W, L, P**) by considering the damage to the road where $\mathbf{A} = \{a_1, a_2, \ldots, a_n\}$ denotes all the rescue road with a_i being the rescue road i; $\mathbf{W} = \{w_1, w_2, \ldots, w_n\}$ where w_i is the width of a_i; $\mathbf{L} = \{l_1, l_2, \ldots, l_n\}$ where w_i is the width of a_i; and $\mathbf{P} = \{p_1, p_2, \ldots, p_n\}$ with p_i being the probability that road a_i would be damaged in an earthquake. For instance, Fig. 1 demonstrates a rescue network with 11 roads in Tainan, Taiwan for the ambulances transporting from the affected area to the CCP and the information, including length, width and road failure is involved in each road a_i.

In this study, we focus on the rescue operation which is the transportation of casualties to the CCP by ambulance. Demand D of the rescue network denotes the number of casualties that must be transported to the CCP. The transport time for the ambulance on each road is denoted as x_i. Due to the damage on each road in an earthquake situation, such as building collapsed or traffic, the ambulance transport time is set as a random integer variable. Therefore, such a rescue network is treated as an SRN and two assumptions are satisfied as follows:

i. The duration of each a_i takes integer values according to a given distribution.

ii. The durations of different a_i are statistically independent.

Note that, assumption *ii* is necessary for calculating the network reliability.

2.2. Transport time estimation of ambulance and time probability table establishment

In a rescue operation, the casualties can be whether successfully treated in time is the most important issue. The transport time of the ambulance is adopted to evaluate the efficiency of rescue. In this section, the method of estimating the transport time of each road is firstly carried out by the traffic flow model. The multiple transport time are then obtained to represent the damage to the rescue road. Finally, the probability would be assigned to each state to generate the stochastic transport time of each road which is presented in the time probability table. To describe the speed of the ambulance in the rescue operation, a concept of Pipes' generalized model [44] is used since further developments were made for adding a new parameter φ to provide a more generalized approach for describing the relationship between the traffic factors, including speed and vehicle density. The average ambulance speed v_i transformed from vehicle density of road i exactly implies

$$v_i = v_f\left[1 - \left(\frac{d}{d_i}\right)^{\varphi}\right] \quad \text{for } i = 1, 2, \cdots, n, \tag{1}$$

where v_f is free-flow speed of ambulance which represents the ambulance speed without influences from other vehicles., d is the pre-set vehicle density, d_i is the jam density of road i and φ ($\varphi \geq 2$) is used to expend the model to be polynomial or quadratic. Note that, the parameters mentioned in Pipes' generalized model are calibrated in different cases and the jam density is mainly chosen from those modified parameters to describe the damage on the rescue network by regarding the transport time of the ambulance as a stochastic integer variable. To represent the stochastic transport time, the jam density under damage is pre-set to represent the damage of the road. The jam density would be adjusted by the probability of road failure p_i, which is used to classify the damage level when the road is damaged in an earthquake. In summary, the jam density of each road would be divided into several states to indicate the damage on each road, including the undamaged and damaged states. The details of classifying the jam density would be introduced in the case study.

For each average ambulance speed calculated by Eq. (1), the transport time of each state can be further transformed as follows,

$$t_i = \frac{l_i}{v_i}. \tag{2}$$

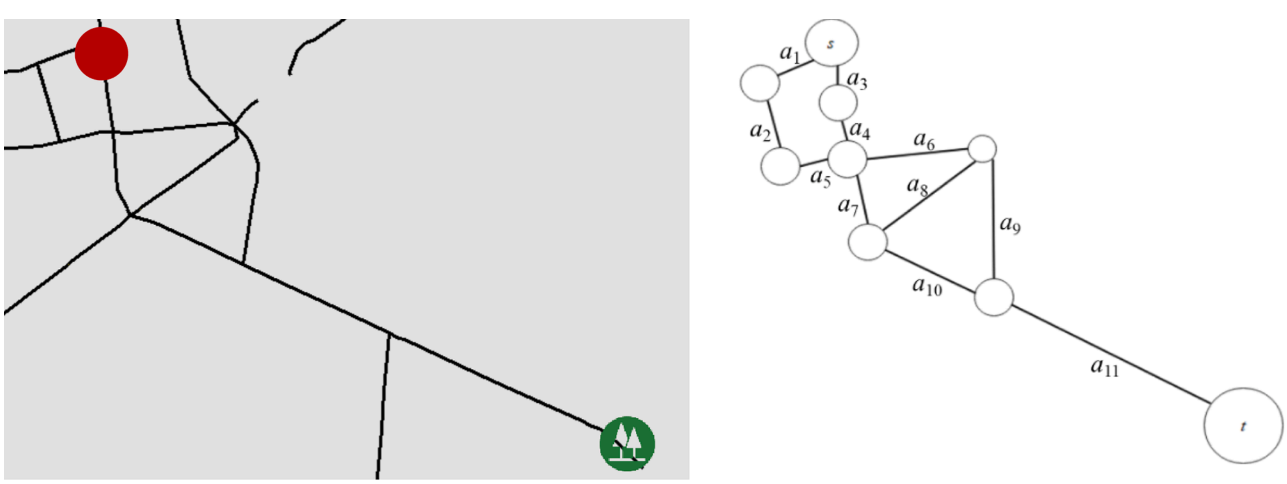

Fig. 1. A rescue network example.

Multiple transport times under different damage t_i^o, t_i^m and t_i^s can be depicted where t_i^o is the transport time in the undamaged state for a_i, t_i^m is the transport time in the moderate state for a_i and t_i^s is the transport time in the severe state for a_i. The corresponding probabilities of each transport time, p_i^o, p_i^m and p_i^s are further determined by the road failure parameter p_i. Specifically, the values of p_i^o is set as (1-p_i), and the value of p_i^m and p_i^s are set as the half value of p_i. It should be noted that the value of p_i is determined by the historical data, and the ratio for setting p_i^m and p_i^s can be adjusted to accurately describe the practical disaster situation by establishing the different time probability table. An example of time probability is shown in Table 1. If we adjust the ratio as 1/5, the corresponding values for p_i^m and p_i^s would be 0.05 and 0.20, respectively. It indicates that 20% probability of severe damage on that particular road, reflecting a large magnitude earthquake.

2.3. Reliability evaluation in terms of maximal duration vector under demand D

To efficiently transport the casualties to the CCP, the concept of MP is used as the rescue route for the ambulance. Let R_j be the jth MP in the SRN for $j = 1, 2, \ldots, m$ where R is a vector of arcs connecting the affected area and CCP without cycles. For instance, in Fig. 1, there are 8 MP as rescue routes for ambulance transporting, $R_1 = (a_2, a_4, a_7, a_8)$, $R_2 = (a_1, a_4, a_7, a_8)$, $R_3 = (a_2, a_4, a_5, a_6, a_8)$, $R_4 = (a_2, a_3, a_6, a_8)$, $R_5 = (a_2, a_3, a_5, a_7, a_8)$, $R_6 = (a_1, a_4, a_5, a_6, a_8)$, $R_7 = (a_1, a_3, a_6, a_8)$, $R_8 = (a_1, a_3, a_5, a_7, a_8)$. Considering the variable condition under the selected rescue routes R_j, the network reliability becomes a critical indicator to evaluate the rescue efficiency. Let $X = (x_1, x_2, \ldots, x_n)$ be the duration vector of the SRN where x_i is the current transport time of a_i. For convenience, let $\mathbf{X_D}$ be the set of duration vectors that can transport the casualties demand D in time and thus the network reliability is $\sum_{X \in \mathbf{X}_D} \Pr\{X\}$. However, when facing a large network, it is an inefficient method by enumerating all $X \in \mathbf{X_D}$ that may increase the computational time. To efficiently calculate the network reliability, the maximal duration vector D-MDV is defined as the maximal required-transport time of an ambulance to transport the number of casualties D.

Definition 1. X is a duration vector with $X \in \mathbf{X_D}$. If any duration vector Y with $Y > X$ such that $Y \notin \mathbf{X_D}$, then X is called a D-MDV (where $Y \geq X$ if and only if $y_i \geq x_i$ for $i = 1, 2, \ldots, n$ and $Y > X$ if and only if $Y \geq X$ and $y_i > x_i$ for at least one i). That is, a D-MDV is the maximal one from $\mathbf{X_D}$.

Suppose $X_1, X_2, \ldots, X_\omega$ are all D-MDVs. The network reliability can be rewritten to $\Pr\{\bigcup_{i=1}^{\omega}(X | X \leq X_i)\}$, and Recursive Sum of Disjoint Product (RSDP$^+$) [45] is one of an efficient method to efficiently calculate such the probability.

All ambulances need to transport the casualties through the given routes R_j. The number of trips for ambulance is firstly computed as below,

$$\eta = \left\lceil \frac{D}{c} \right\rceil 2 - 1. \tag{4}$$

where $\lceil D/c \rceil$ is the required number of trips for the available ambulances c to transport all the casualties D from the affected area to hospital. However, the ambulances must return to the affected area to transport the remaining casualties, except for the last trip. To ensure that all casualties can be transported by the given number of ambulances, the remaining casualties can be rescued on the last trip by adopting the concept of ceiling function.

The average transport time θ for each trip can be further obtained by $\theta = \lfloor T/\eta \rfloor$. Note that, the travel time of each arc is assumed to be an integer in our study, which means the average transport time θ must be calculated using the floor function to make sure the total transport time would not be exceeded. For each R_j, a pseudo maximal duration vector $Z = (z_1, z_2, \ldots, z_n)$ satisfying θ is computed by

$$\sum z_i = \theta, \text{if} \quad a_i \in R_j \text{ and } \quad t_i^s, \text{if} \quad a_i \notin R_j \qquad \text{for} \quad i = 1, 2, \ldots, n, \tag{5}$$

$$z_i \geq t_i^o \quad \text{for } i = 1, 2, \cdots, n. \tag{6}$$

where $\sum_{a_i \in R_j} z_i$ represents the summation of transport time of a_i in R_j. Eq. (5) denotes that the transport time of ambulance in R_j equal to the average transport time for each trip, and the maximal transport time is set for the unused road. Eq. (6) represents that the transport time cannot smaller than the minimal transport time for a_i. For convenience, let $\mathbf{Z_\theta}$ be the set of pseudo maximal duration vector satisfying Eqs. (5)-(6) to meet the average transport time θ for each trip. That is, the duration vector satisfying D can be represented as a Z satisfying θ. In order to obtain all $Z \in \mathbf{Z_\theta}$, several methods can be applied such as branch-and-bound [46] and backtracking [47] methods. For example, a rescue operation involving three casualties and one ambulance with the rescue route $R = (a_2, a_4, a_7, a_{10}, a_{11})$ depicted in Fig. 1. By Eq. (4), the number of trips for ambulance η can be obtained through $\lceil 3/1 \rceil . \times 2 + 1 = 7$. If we set the rescue time at 21 minutes, the average transport time θ can be calculated by $\lfloor 21/7 \rfloor = 3$, indicating that the transport time for each ambulance trip is up to three minutes. The pseudo maximal vectors can be further obtained by finding all combinations such that $z_2 + z_4 + z_7 + z_{10} + z_{11} = 3$ where the travel time of the other unpassed road is set as the maximal travel time for each arc. By Eqs. (5)-(6), the maximal travel time of each road under rescue time T can be obtained in terms of all Z. Suppose that there are v pseudo maximal duration vectors: $Z_1, Z_2, \ldots, Z_v$. A group of $X = (x_1, x_2, \ldots, x_n)$ corresponding to each $Z \in \mathbf{Z_\theta}$ is transformed as follows,

$$x_i = \begin{cases} t_i^o, \text{if} t_i^o \leq z_i \leq t_i^m \\ t_i^m, \text{if} t_i^m \leq z_i \leq t_i^s & \text{for } i = 1, 2, \ldots, n \\ t_i^s, other \end{cases} \tag{7}$$

The following lemma shows a critical property of D-MDV.

Lemma 1. If X is a D-MDV, then there exists at least one pseudo maximal duration vector $Z \in \mathbf{Z_\theta}$ such that the X is transformed from such a Z via Eq. (7).

Proof. Suppose X is a D-MDV, then $X \in \mathbf{X_D}$ such that $\sum_{a_i \in R_j} x_i \leq \theta$. If $\sum_{a_i \in R_j} x_i = \theta$, then set $Z = X$. Otherwise, set $X^1 = X + e = (x_1^1, x_2^1, \ldots, x_n^1)$ where e is the time unit added in any one of the arcs whose capacity is not the maximal. If $\sum_{a_i \in R_j} x_i^1 = \theta$ then Z is regarded as X^1 which is belonging to $\mathbf{Z_\theta}$. Otherwise, the same procedure can be repeated for X^1 in finite steps. Specifically, an integer τ is existed where the comparison procedures with τ times can be carried out such that $X^\tau > X^{\tau\text{-}1} > \ldots > X^1 > X$ with $\sum_{a_i \in R_j} x_i^\tau = \theta$ and $\sum_{a_i \in R_j} x_i^{\tau+1} > \theta$. Therefore, the pseudo maximal duration vector Z can be obtained by letting $Z = X_\tau$. That is, $Z \geq X$ and $Z \in \mathbf{Z_\theta}$. Moreover, suppose to the contrary that there exists a duration vector Y with $\sum_{a_i \in R_j} y_i \leq \theta$ such that $Z \geq Y > X$. Then $Y \in \mathbf{X_D}$ which is contrary to X is a D-MDV. Thus, for any D-MDV we can find the corresponding $Z \in \mathbf{Z_\theta}$ transformed via Eq. (7).

Let $\mathbf{\Omega} \equiv \{X \mid X$ is transformed from an $Z \in \mathbf{Z_\theta}$ by Eq. (7)$\}$ and $\mathbf{\Omega_{max}} \equiv \{X \mid X$ is a maximal duration vector in $\mathbf{\Omega}\}$. According to Lemma 1, the set $\mathbf{\Omega}$ contains all D-MDVs and thus each $X \in \mathbf{\Omega}$ is called a D-MDV candidate. The following Lemma further shows that all D-MDV is exactly $\mathbf{\Omega_{max}}$.

Lemma 2. $\mathbf{\Omega_{max}}$ is the set of D-MDVs.

Table 1
The time probability example with $p_i = 0.25$.

	Transport time* (t_i)	Probability (p_i)
a_i	20 (t_i^o)	0.750 (p_i^o)
	32 (t_i^m)	0.125 (p_i^m)
	47 (t_i^s)	0.125 (p_i^s)

* unit: mins

Proof. Let $X \in \mathbf{\Omega}_{\mathbf{max}}$ (note that $X \in \mathbf{\Omega}$) but it is not a D-MDV. Then a D-MDV Y in $\mathbf{\Omega}$ would exist such that $Y > X$ by Lemma 1. It infers that $Y \in \mathbf{X}_D$ which contradicts to the $X \in \mathbf{\Omega}_{\mathbf{max}}$. Therefore, X is a D-MDV. In contrast, suppose the duration vector X is a D-MDV ($X \in \mathbf{\Omega}$ by Lemma 1) but $X \notin \mathbf{\Omega}_{\mathbf{max}}$ i.e., there exists a $Y \in \mathbf{\Omega}_{\mathbf{max}}$ such that $Y > X$. It implies that $Y \in \mathbf{X}_D$ which contradicts X is a D-MDV. Therefore, $X \in \mathbf{\Omega}_{\mathbf{max}}$.

Overall, we learn that the set $\mathbf{\Omega}$ obtained from Eq. (7) includes all D-MDVs and the set $\mathbf{\Omega}_{\mathbf{max}}$ of maximal duration vectors can be obtained by removing the non-maximal ones from $\mathbf{\Omega}$ and thus is also belonging to $\mathbf{\Omega}$. The following comparison procedure is carried out to obtain all D-MDVs from $\mathbf{\Omega}$.

The RSDP^{+} method is then adopted to calculate the network reliability by $\mathbf{\Omega}_{\mathbf{max}}$ as Procedure 2.

The whole procedure is presented by the flow chart in Fig. 2.

3. Proposed algorithm

In this section, the whole procedure for establishing the time probability table and calculating the network reliability in terms of all D-MDVs is presented.

3.1. Algorithm for evaluating network reliability of an SRN

INPUT: rescue network (**A, W, L, P**), D, c, T

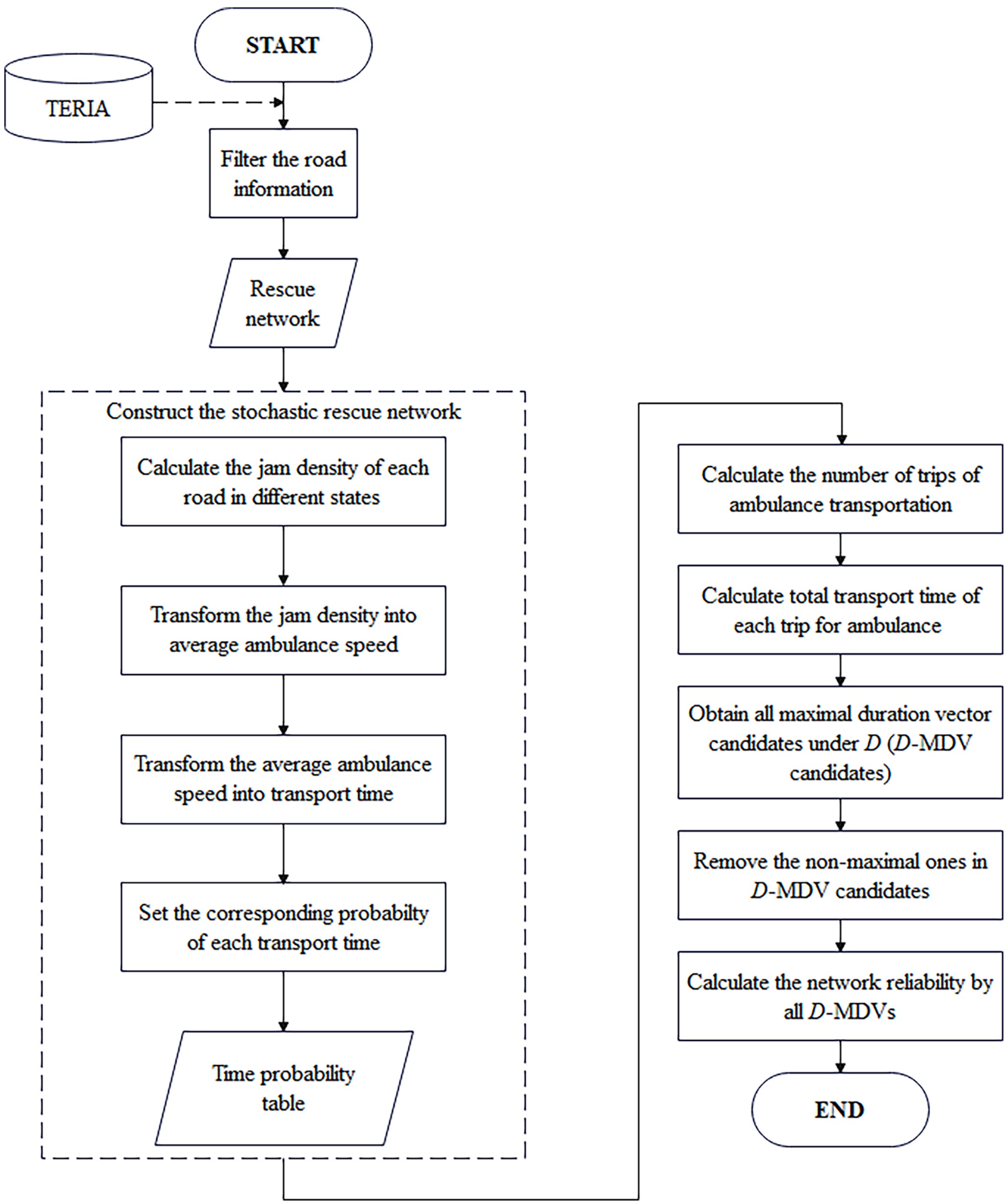

Fig. 2. The whole procedure for evaluating the network reliability.

Step 1. Transform the average ambulance speed from jam density under different states for each arc according to the following equation.

$$v_i = v_f\left[1-\left(\frac{d}{d_i}\right)^{\varphi}\right] \text{for } i = 1, 2, \cdots, n. \tag{8}$$

Step 2. Compute the transport time from the length and average ambulance speed obtained by Step 1 for each arc.

$$t_i = \frac{l_i}{v_i}. \tag{9}$$

Step 3. Calculate the average transport time for each trip.

3.1) Employ the following equation to obtain the number of trips for ambulance transporting η.

$$\eta = \left\lceil \frac{D}{c} \right\rceil \times 2 - 1. \tag{10}$$

3.2) Use the following equation to compute θ.

$$\theta = \left\lfloor \frac{T}{\eta} \right\rfloor. \tag{11}$$

Step 4. Obtain pseudo maximal duration vector Z using the following equation,

$$\begin{matrix} \sum z_i = \theta, \text{if} \quad a_i \in R_j \text{ and} \\ t_i^s, \text{if} \quad a_i \notin R_j \end{matrix} \quad \text{for} \quad i = 1, 2, ..., n, \tag{12}$$

$$z_i \geq t_i^o. \tag{13}$$

Step 5. Obtain D-MDV candidates X from Step 4 using the following equation,

$$x_i = \begin{cases} t_i^o, \text{if} t_i^o \leq z_i \leq t_i^m \\ t_i^m, \text{if} t_i^m \leq z_i \leq t_i^s, for\ i = 1, 2, ..., n. \\ t_i^s, other \end{cases} \tag{14}$$

Step 6. Suppose $X_1, X_2, \ldots, X_{|\Omega|}$ are the D-MDV candidates. Use Procedure 1 to obtain all D-MDVs.

Step 7. Suppose $X_1, X_2, \ldots, X_{|\Omega_{\max}|}$ are all D-MDVs. Use Procedure 2 to calculate the network reliability R_D.

Output: R_D

3.2. Computational complexity analysis

In the proposed algorithm, the number of solutions to Eqs. (12)-(13) is $\lambda \equiv \binom{n+\theta-1}{\theta}$ in the worst case, which means that all number of arcs may be included in one MP.

Computational complexity of steps 1 and 2:

Each arc contains 3 states. Each solution of Eq. (8) thus needs $O(1)$ time to obtain the stochastic average ambulance speed for arc i, and at most $O(n)$ for all arcs.

Computational complexity of step 3:

For step 3, the number of trips and average transport time under given time constraint are obtained. It takes $O(1)$ time to obtain those two parameters and is not affected by the number of arcs.

Computational complexity of step 4:

Each solution of Eqs. (12)-(13) needs at most $O(n)$ time to examine whether it meets $t_i^s \leq z_i$ for each i. Therefore, step 4 takes $O(n\lambda)$ time to obtain all feasible pseudo vectors Z in the worst case.

Computational complexity of step 5:

Each pseudo vector needs to take $O(n)$ time to transform into the maximal duration vector. Therefore, step 5 takes $O(n\lambda)$ time to obtain all maximal duration vector candidates in the worst case as the number of D-MDV is $|\mathbf{\Omega}| = \lambda$.

Computational complexity of step 6:

Step 6 includes executing Procedure 1 to remove the non-maximal ones in $\mathbf{\Omega}$. Procedure 1 takes at most $O(n\lambda)$ time to test one candidate whether it is greater than another or not, and thus $O(n\lambda^2)$ for all candidates. In the worst cast, step 6 takes $O(n\lambda^2)$ time to implement Procedure 1.

Computational complexity of step 7:

Step 7 involves implementing Procedure 2 which takes $O(n)$ time to execute the RSDP function for each D-MDV in the worst case. Therefore, it takes $O(n\lambda)$ in the worst case as the number of D-MDV is $|\mathbf{\Omega}| = \lambda$.

In summary, the proposed algorithm needs $O(n^2\lambda)$ time in the worst case and it is sensitive to average transport time θ, which is transformed by demand D, transport time constraint T and number of ambulance c.

4. Case study of a rescue operation in Taiwan

In Taiwan, Tainan County is often severely damaged by an earthquake due to the property of the weak formation. Many historical earthquakes have occurred around Tainan and cause serious injuries. In a rescue operation, it is important for the government to calculate the network reliability to assess whether the ambulance allocation is assigned properly. The historical earthquake in Tainan on February 6, 2016 is adopted to analyze the rescue efficiency of the entire rescue operation. More than one hundred people died and five hundred people were injured in this incident. Rescue becomes more difficult as there are also more than 10,000 buildings affected. The government implements a rescue plan that a group of casualties must be transported to the CCP in Yongkang, Tainan (see Fig. 3). Since there are multiple roads that can connect to the same destination, the different roads are merged into a_i to represent the connection of two areas. For example, a total of k roads located in the same area i are emerged to an arc a_i with minimal width and maximal road failure (see Fig. 4).

According to the Regional Disaster Prevention and Rescue Plan in Tainan City, the rescue roads are selected based on the value of road failure. The road would not be adopted as the rescue road when p_i is larger than 0.5 and w_i is less than 8 meters, and the probability p_i is regarded as 0.1 while $p_i < 0.1$ in this case in order to show the damage of the road. Also, the undamaged jam density of road i is determined by the width. Table 2 shows the standard of jam density without damage. In an earthquake situation, the other two states are carried out since each road may be damaged and cause the jam density decrease. Table 3 shows the decrease level of jam density set in this case. The parameter for transforming the ambulance speed d is set as 180 number of vehicles/kilometer and v_f is set as 30 kilometer/hour.

4.1. Demonstration for calculating network reliability in Yongkang district

Under one rescue operation with $D = 8$, Fig. 5 demonstrates a rescue network after filtering and merging the rescue road. Five MPs with minimal transport time are adopted as rescue route, $R_1 = (a_4, a_6, a_6, a_{15}, a_{20}, a_{27})$, $R_2 = (a_2, a_3, a_5, a_6, a_{10}, a_{15}, a_{20}, a_{27})$, $R_3 = (a_1, a_5, a_6, a_{10}, a_{15}, a_{20}, a_{27})$, $R_4 = (a_4, a_6, a_{10}, a_{15}, a_{16}, a_{17}, a_{22}, a_{27})$, $R_5 = (a_4, a_6, a_{10}, a_{15}, a_{16}, a_{21}, a_{27})$. The related information of each arc is shown as Table 4. By the proposed algorithm, RSS can easily evaluate the rescue efficiency and also determine the allocation of the ambulances. The network reliability calculation is performed by the following steps. The proposed algorithm is programmed with Python programming language and

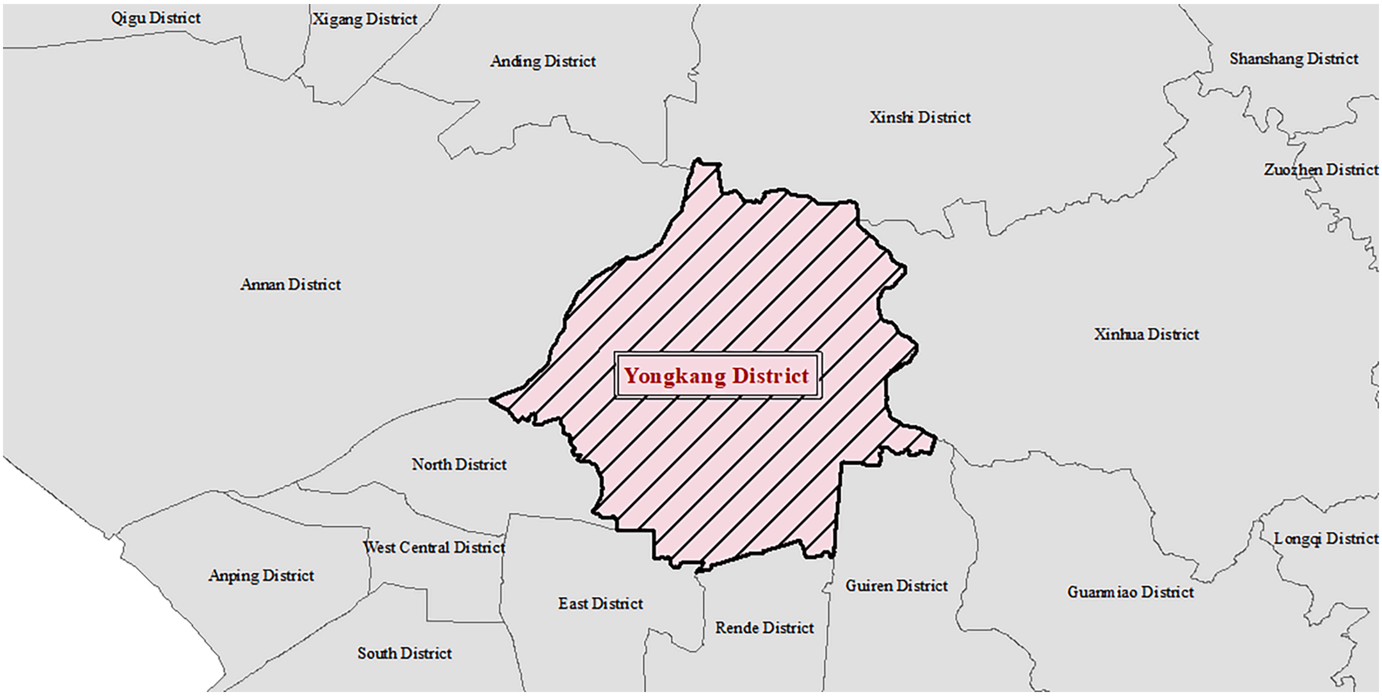

Fig. 3. The studied area in Tainan City.

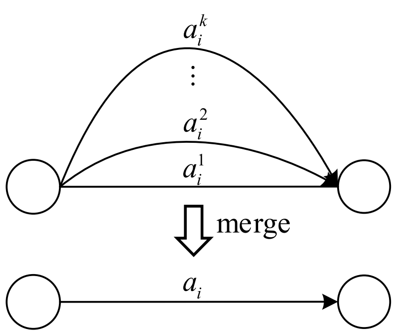

Fig. 4. Concept of merging rescue road.

Table 2
Standard of jam density level in undamaged state.

road width	jam density of undamaged state*
over 20 meters	400
15 – 19 meters	300
8 – 14 meters	200

* unit: number of vehicle/kilometer

Table 3
Standard of jam density level in moderate and severe states.

road state	jam density*
undamaged	d_i^o
moderate	$\frac{3}{4}d_i^o$
severe	$\frac{1}{2}d_i^o$

* unit: number of vehicle/kilometer

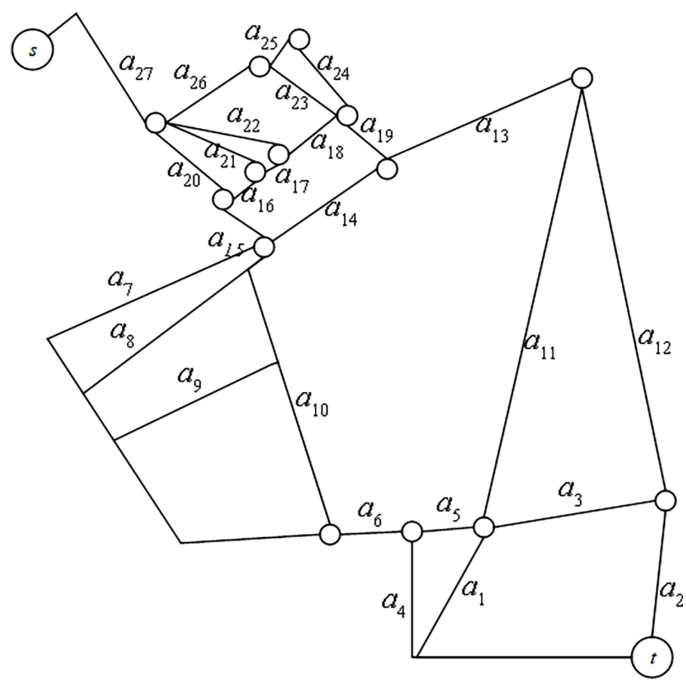

Fig. 5. The network topology in Yongkang district.

executed on a personal computer with Intel® Core^{TM} i9-9900K CPU 3.60 GHz and 32 GB of RAM. In this section, we demonstrate the rescue operation with 2 MP: R_3, R_4 to transport the casualties to the CCP. Firstly, the jam density of three states for each arc (See Table 5) should be obtained by the standard in Tables 2 and 3.

INPUT: rescue network (**A, W, L, P**) (Tables 4, 5), $D = 8$, $c = 3$, $T = 435$

Step 1. Transform the average ambulance speed from jam density under different states for each arc, such that

Table 4
Road information in Yongkang district.

a_i	Length* (l_i)	Width* (w_i)	Road failure (p_i)
a_1	927.299	11	0.005238
a_2	459.002	13	0.000192
a_3	350.965	10	0.005238
a_4	858.851	9	0.005238
a_5	187.221	11	0.005238
a_6	189.032	8	0.005238
a_7	1393.187	12	0.5
a_8	1316.709	9	0.5
a_9	1346.104	8	0.161035
a_{10}	790.052	8	0.005238
a_{11}	1113.016	11	0.005238
a_{12}	1222.152	12	0.005238
a_{13}	492.289	18	0.005238
a_{14}	364.001	19	0.000192
a_{15}	189.711	13	0.000192
a_{16}	122.146	18	0.000192
a_{17}	73.477	18	0.000192
a_{18}	162.051	19	0.000192
a_{19}	73.693	20	0.000192
a_{20}	253.278	20	0.000192
a_{21}	388.948	10	0.005238
a_{22}	254.323	10	0.005238
a_{23}	334.79	15	0.000192
a_{24}	99.041	20	0.000192
a_{25}	118.359	9	0.005238
a_{26}	321.832	10	0.005238
a_{27}	365.72	12	0.000192

* unit: meter

Table 5
Jam density of different states in each road.

a_i	jam density*		
	undamaged	moderate	severe
a_1	200	150	100
a_2	200	150	100
a_3	200	150	100
a_4	200	150	100
a_5	200	150	100
a_6	200	150	100
a_7	200	150	100
a_8	200	150	100
a_9	200	150	100
a_{10}	200	150	100
a_{11}	200	150	100
a_{12}	200	150	100
a_{13}	300	225	150
a_{14}	300	225	150
a_{15}	200	150	100
a_{16}	300	225	150
a_{17}	300	225	150
a_{18}	300	225	150
a_{19}	400	300	200
a_{20}	400	300	200
a_{21}	200	150	100
a_{22}	200	150	100
a_{23}	300	225	150
a_{24}	400	300	200
a_{25}	200	150	100
a_{26}	200	150	100
a_{27}	200	150	100

* unit: number of vehicle/kilometer

Table 6
The result from Step 1.

a_i	average speed*		
	undamaged	moderate	severe
a_1	12.19709	9.035826	4.958967
a_2	12.19709	9.035826	4.958967
a_3	12.19709	9.035826	4.958967
a_4	12.19709	9.035826	4.958967
a_5	12.19709	9.035826	4.958967
a_6	12.19709	9.035826	4.958967
a_7	12.19709	9.035826	4.958967
a_8	12.19709	9.035826	4.958967
a_9	12.19709	9.035826	4.958967
a_{10}	12.19709	9.035826	4.958967
a_{11}	12.19709	9.035826	4.958967
a_{12}	12.19709	9.035826	4.958967
a_{13}	16.46435	13.47987	9.035826
a_{14}	16.46435	13.47987	9.035826
a_{15}	12.19709	9.035826	4.958967
a_{16}	16.46435	13.47987	9.035826
a_{17}	16.46435	13.47987	9.035826
a_{18}	16.46435	13.47987	9.035826
a_{19}	19.12884	16.46435	12.19709
a_{20}	19.12884	16.46435	12.19709
a_{21}	12.19709	9.035826	4.958967
a_{22}	12.19709	9.035826	4.958967
a_{23}	16.46435	13.47987	9.035826
a_{24}	19.12884	16.46435	12.19709
a_{25}	12.19709	9.035826	4.958967
a_{26}	12.19709	9.035826	4.958967
a_{27}	12.19709	9.035826	4.958967

* unit: kilometers/hour

$$v_i = v_f\left[1-\left(\frac{d}{d_i}\right)^{\varphi}\right] \quad \text{for } i = 1, 2, \cdots, n. \tag{15}$$

Take a_1 for example, the average ambulance speed in undamaged state is $v_1^o = 30[1\text{-}(180/200)^2] = 12.19709$. From Step 1, all of the average speed for each state are obtained, and the whole results are presented as Table 6. Note that, the unit of road length in Table 4 would be adjusted to kilometer before proceeding with Step 1 for calculating the ambulance speed.

Step 2. Compute the transport time from the length and average ambulance speed obtained by Step 1 for each arc. Subsequently, the corresponding probability for each state is set up to represent the stochastic transport time under earthquake. Take a_1 in undamaged state for example, the transport time $t_1^o = 0.927299/12.19709 = 273.694496$ (sec). The final result for representing the time probability table is shown as Table 7.
Step 3. Calculate the average transport time for each trip.
3.1) The number of trips for ambulance transporting is $\eta = \lceil 8/3 \rceil \times 2 - 1 = 5$.
3.2) The average transport time for each trip is $\theta = \lfloor 435/5 \rfloor = 87$.
Step 4. Obtain all pseudo maximal duration vector Z for each route satisfying the following constraints. Take a_1 in R_3 for example:

$$\sum_{a_i \in R_3} z_i = z_1 + z_5 + z_6 + z_{10} + z_{15} + z_{20} + z_{27} = 87 \tag{16}$$

The other pseudo maximal durations which are not belonged to R_3 are regarded as their maximal transport time. The result is presented in Table 8

Step 5. From Step 4, there are 15 pseudo maximal duration vectors. Each vector would be transformed to a D-MDV candidate. Take a_1 in

Table 7
Time probability table for each arc.

a_i	t*	probability	a_i	t*	probability	a_i	t*	probability
a_1	27.36945	0.9	a_{10}	23.31857	0.9	a_{19}	1.386884	0.9
	28.17569	0.05		24.00548	0.05		1.407163	0.05
	29.06188	0.05		24.76051	0.05		1.429121	0.05
a_2	13.54754	0.9	a_{11}	32.85092	0.9	a_{20}	4.766621	0.9
	13.94662	0.05		33.81864	0.05		4.836319	0.05
	14.38527	0.05		34.88231	0.05		4.911787	0.05
a_3	10.35882	0.9	a_{12}	36.07211	0.9	a_{21}	11.47989	0.9
	10.66396	0.05		37.13472	0.05		11.81807	0.05
	10.99937	0.05		38.30269	0.05		12.18977	0.05
a_4	25.34918	0.9	a_{13}	13.65697	0.9	a_{22}	7.506413	0.9
	26.09592	0.05		13.92387	0.05		7.727536	0.05
	26.91669	0.05		14.21432	0.05		7.970584	0.05
a_5	5.525863	0.9	a_{14}	7.959026	0.9	a_{23}	7.320331	0.9
	5.688643	0.05		8.114572	0.05		7.463394	0.05
	5.867564	0.05		8.28384	0.05		7.619079	0.05
a_6	5.579319	0.9	a_{15}	5.59938	0.9	a_{24}	1.863927	0.9
	5.743673	0.05		5.764325	0.05		1.891181	0.05
	5.924325	0.05		5.945626	0.05		1.920692	0.05
a_7	41.12025	0.5	a_{16}	2.670778	0.9	a_{25}	3.493399	0.9
	55.50653	0.25		2.722974	0.05		3.596307	0.05
	101.1395	0.25		2.779774	0.05		3.709419	0.05
a_8	38.86297	0.5	a_{17}	1.606599	0.9	a_{26}	9.498955	0.9
	52.45953	0.25		1.637997	0.05		9.778773	0.05
	95.58749	0.25		1.672165	0.05		10.08634	0.05
a_9	39.73059	0.75	a_{18}	3.543316	0.9	a_{27}	10.79432	0.9
	42.18739	0.25		3.612564	0.05		11.11229	0.05
	45.18173	0.25		3.687921	0.05		11.4618	0.05

* unit: 10 seconds

Z_1 for example, $x_1 = 28$ since 28 (t_i^m) $\le$ 28 (z_1) $<$ 29 (t_i^s). All the D-MDV candidates are shown in the second row of Table 8.
Step 6. Use the Procedure 1 to obtain all *D*-MDVs by comparing all *D*-MDV candidates: X_1, X_2, …, X_{15}. There is total 5 *D*-MDVs and the result is displayed in the third row of Table 8.
Step 7. Use Procedure 2 to calculate the network reliability R_D from the *D*-MDVs: R_8 Pr$\{\bigcup_{i=1}^{5}(X|X \le X_i)\}$= 0.9882.

Output: $R_8 = 0.9882$

4.2. Numerical experiments

This study examines the use of network reliability in rescue operations and how it can build an RSS in evaluating the extent of damage to rescue network and allocating ambulances to transport casualties to the CCP. In this section, the impact of vary rescue times and number of routes on network reliability is discussed to provide recommendations on rescue operations. For a given rescue network used to transport eight casualties (see Fig. 5), rescue times ranging from 3900 to 4600 seconds are considered for five different rescue routes: R_1, R_2, R_3, R_4 and R_5. Table 9 summarizes the experimental results, which show an upward trend in network reliability as the rescue time increases. All information in Table 9 calculated in real time is useful for developing an RSS when determining a rescue route or rescue time. For example, the rescue route should be set as R_1 or R_2 to ensure the rescue time is not too long. Under the fixed rescue route R_2, the rescue time should not exceed 4200 seconds to ensure that the network reliability is greater than 0.9. It is worth noting that the network reliability of choosing R_1 is higher because the average transport time of R_1 is the lowest. By quantifying the rescue efficiency, it becomes possible to comprehensively assess the risk associated with each ambulance assignment. The results of Table 9 aids the rescue commander or ambulance driver in making informed decisions regarding the optimal rescue route and even determining the necessity of additional ambulances.

This study also takes the number of routes into consideration and the results are listed in Table 10. For a rescue time of 4,300 seconds, choosing R_3 and R_4 results in higher network reliability than selecting only R_3 or the combination of R_3, R_5. An RSS can use this information to easily select the best combination of rescue routes. Furthermore, the results for (R_3, R_4) compared to (R_3, R_4, R_5) are the same, indicating that the rescue route R_5 is unnecessary for transporting casualties. To easily observe the relationship between rescue time and network reliability, Fig. 6 is presented for different route selections. The figure shows that network reliability increases with increased rescue time and number of routes selected. Based on this indicator, An RSS can provide the suggestion for assigning the alternative rescue route for the ambulance. For example, if the rescue time is set to be 4325 seconds, an ambulance should transport the casualties via R_4, and R_3 should be considered an alternative rescue route. In summary, the experimental results can be totally entering to an RSS to provide the main and alternative rescue route for each ambulance. Under more rescue operation are required simultaneously for a rescue commander, the rescue time for each transport can be determined based on the recommendations by RSS. In case of unexpected congestion or damage along the rescue route, alternative routes can be suggested to make sure the timely arrival of casualties at the CCP.

5. Discussion and Conclusion

In an RSS, it is important to provide the commander with rescue suggestions, such as the dispatching of rescue routes. Therefore, the concept of expected time can be used and calculated by the results in Tables 9 and 10. The expected transport time for a particular route R, denoted as E_R, can be calculated as follows: take R_1 for example, E_{R_1} = 3900 × 0.729 + 3950 × (0.972-0.729) + 4000 × (0.999-0.972) + 4050 × (1-0.999) = 3915. Such value means that an RSS can show the transport time of ambulance through R_1 with an approximate of 65 minutes. The expected transported time is calculated by the summation of all products of the rescue time and network reliability, and the expected time for the other routes can be also calculated by this method and is shown in Table 11. Note that, before calculating the expected transport time, the results of Tables 9 and 10 should be obtained firstly by the proposed approach. According to Table 11, we can conclude that R_1 and R_2 are the most efficient solutions for ambulance transport. For

Table 8
The result from Steps 4, 5 and 6.

Step 4	Step 5	Step 6
$Z_1 = (28, 15, 11, 27, 6, 6, 102, 96, 46, 24, 35, 39, 15, 9, 6, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 12)$	$X_1 = (28, 15, 11, 27, 6, 6, 102, 96, 46, 24, 35, 39, 15, 9, 6, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 12)$	X_1 is a *D*-MDV.
$Z_2 = (29, 15, 11, 27, 6, 6, 102, 96, 46, 24, 35, 39, 15, 9, 6, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 11)$	$X_2 = (29, 15, 11, 27, 6, 6, 102, 96, 46, 24, 35, 39, 15, 9, 6, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 11)$	No, $X_2 < X_1$
$Z_3 = (28, 15, 11, 27, 7, 6, 102, 96, 46, 24, 35, 39, 15, 9, 6, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 11)$	$X_3 = (28, 15, 11, 27, 6, 6, 102, 96, 46, 24, 35, 39, 15, 9, 6, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 11)$	X_3 is a *D*-MDV.
$Z_4 = (28, 15, 11, 27, 6, 7, 102, 96, 46, 24, 35, 39, 15, 9, 6, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 11)$	$X_4 = (28, 15, 11, 27, 6, 6, 102, 96, 46, 24, 35, 39, 15, 9, 6, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 11)$	No, $X_4 = X_3$
$Z_5 = (28, 15, 11, 27, 6, 6, 102, 96, 46, 25, 35, 39, 15, 9, 6, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 11)$	$X_5 = (28, 15, 11, 27, 6, 6, 102, 96, 46, 25, 35, 39, 15, 9, 6, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 11)$	No, $X_5 = X_3$
$Z_6 = (28, 15, 11, 27, 6, 6, 102, 96, 46, 24, 35, 39, 15, 9, 7, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 11)$	$X_6 = (28, 15, 11, 27, 6, 6, 102, 96, 46, 24, 35, 39, 15, 9, 6, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 11)$	No, $X_6 = X_3$
$Z_7 = (28, 15, 11, 27, 6, 6, 102, 96, 46, 24, 35, 39, 15, 9, 6, 3, 2, 4, 2, 6, 13, 8, 8, 2, 4, 11, 11)$	$X_7 = (28, 15, 11, 27, 6, 6, 102, 96, 46, 24, 35, 39, 15, 9, 6, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 11)$	No, $X_7 = X_3$
$Z_8 = (30, 15, 11, 26, 6, 6, 102, 96, 46, 24, 35, 9, 15, 9, 6, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 12)$	$X_8 = (30, 15, 11, 26, 6, 6, 102, 96, 46, 24, 35, 39, 15, 9, 6, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 12)$	X_8 is a *D*-MDV.
$Z_9 = (30, 15, 11, 27, 6, 6, 102, 96, 46, 24, 35, 39, 15, 9, 6, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 11)$	$X_9 = (30, 15, 11, 27, 6, 6, 102, 96, 46, 24, 35, 39, 15, 9, 6, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 11)$	X_9 is a *D*-MDV.
$Z_{10} = (30, 15, 11, 26, 6, 7, 102, 96, 46, 24, 35, 39, 15, 9, 6, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 11)$	$X_{10} = (30, 15, 11, 26, 6, 6, 102, 96, 46, 24, 35, 39, 15, 9, 6, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 11)$	No, $X_{10} < X_{11}$
$Z_{11} = (30, 15, 11, 26, 6, 6, 102, 96, 46, 25, 35, 39, 15, 9, 6, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 11)$	$X_{11} = (30, 15, 11, 26, 6, 6, 102, 96, 46, 25, 35, 39, 15, 9, 6, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 11)$	X_{11} is a *D*-MDV.
$Z_{12} = (30, 15, 11, 26, 6, 6, 102, 96, 46, 24, 35, 39, 15, 9, 7, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 11)$	$X_{12} = (30, 15, 11, 26, 6, 6, 102, 96, 46, 24, 35, 39, 15, 9, 6, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 11)$	No, $X_{12} < X_{11}$
$Z_{13} = (30, 15, 11, 26, 6, 6, 102, 96, 46, 24, 35, 39, 15, 9, 6, 4, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 11)$	$X_{13} = (30, 15, 11, 26, 6, 6, 102, 96, 46, 24, 35, 39, 15, 9, 6, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 11)$	No, $X_{13} < X_{11}$
$Z_{14} = (30, 15, 11, 26, 6, 6, 102, 96, 46, 24, 35, 39, 15, 9, 6, 3, 3, 4, 2, 5, 13, 8, 8, 2, 4, 11, 11)$	$X_{14} = (30, 15, 11, 26, 6, 6, 102, 96, 46, 24, 35, 39, 15, 9, 6, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 11)$	No, $X_{14} < X_{11}$
$Z_{15} = (30, 15, 11, 26, 6, 6, 102, 96, 46, 24, 35, 39, 15, 9, 6, 3, 2, 4, 2, 5, 13, 9, 8, 2, 4, 11, 11)$	$X_{15} = (30, 15, 11, 26, 6, 6, 102, 96, 46, 24, 35, 39, 15, 9, 6, 3, 2, 4, 2, 5, 13, 8, 8, 2, 4, 11, 11)$	No, $X_{14} < X_{11}$

Table 9
Experiments for different rescue time and route.

R_j / T *	$R_1 = (a_4, a_6, a_6, a_{15}, a_{20}, a_{27})$	$R_2 = (a_2, a_3, a_5, a_6, a_{10}, a_{15}, a_{20}, a_{27})$	$R_3 = (a_1, a_5, a_6, a_{10}, a_{15}, a_{20}, a_{27})$	$R_4 = (a_4, a_6, a_{10}, a_{15}, a_{16}, a_{17}, a_{22}, a_{27})$	$R_5 = (a_4, a_6, a_{10}, a_{15}, a_{16}, a_{21}, a_{27})$
3900	0.7290	0.0000	0.0000	0.0000	0.0000
3950	0.9720	0.0000	0.0000	0.0000	0.0000
4000	0.9990	0.0000	0.0000	0.0000	0.0000
4050	1.0000	0.0000	0.0000	0.0000	0.0000
4100	1.0000	0.0000	0.0000	0.0000	0.0000
4150	1.0000	0.7695	0.0000	0.0000	0.0000
4200	1.0000	0.9810	0.0000	0.0000	0.0000
4250	1.0000	0.9995	0.0000	0.0000	0.0000
4300	1.0000	1.0000	0.7290	0.7290	0.0000
4350	1.0000	1.0000	0.9320	0.9720	0.0000
4400	1.0000	1.0000	0.9900	0.9990	0.6926
4450	1.0000	1.0000	0.9995	1.0000	0.9599
4500	1.0000	1.0000	1.0000	1.0000	0.9977
4550	1.0000	1.0000	1.0000	1.0000	0.9999
4600	1.0000	1.0000	1.0000	1.0000	1.0000

* unit: second

Table 10
Experiments for different combination of rescue route.

R_j	T* 4300	4350	4400	4450	4500	4550	4600
R_3, R_4	0.8019	0.9882	0.9999	1.0000	1.0000	1.0000	1.0000
R_3, R_5	0.7290	0.9315	0.9900	0.9995	1.0000	1.0000	1.0000
R_4, R_5	0.7290	0.9720	0.9990	1.0000	1.0000	1.0000	1.0000
R_3, R_4, R_5	0.8019	0.9882	0.9999	1.0000	1.0000	1.0000	1.0000

* unit: second

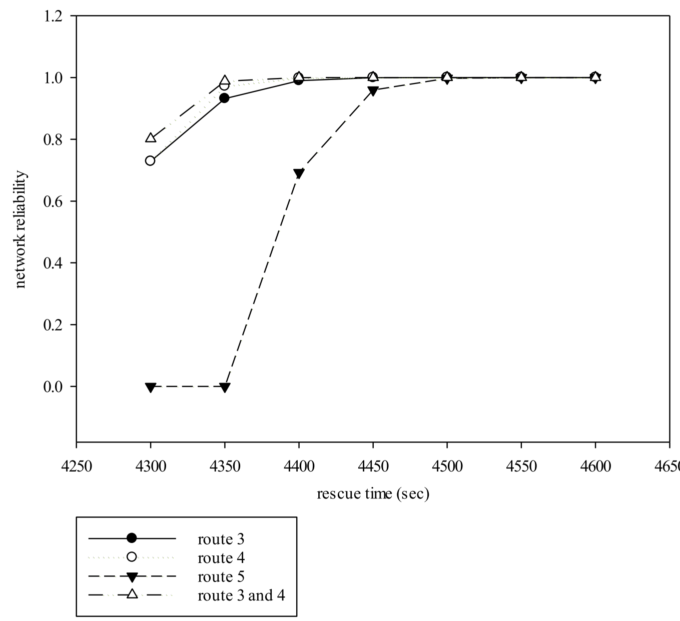

Fig. 6. The relationship between the rescue time and network reliability.

Table 11
Experiments for different combination of rescue route.

rescue route	expected transport time*
R_1	3915.00
R_2	4162.50
R_3	4317.50
R_4	4315.00
R_5	4417.04
R_3, R_4	4310.50
R_3, R_5	4317.50
R_4, R_5	4315.00
R_3, R_4, R_5	4310.50

* unit: second

Procedure 1
The comparison procedure for generating all *D*-MDVs.

INPUT $\mathbf{\Omega} = \{X_1, X_2, \ldots, X_{|\Omega|}\}$ // $|\Omega|$ is the number of *D*-MDV candidates
Set $\mathbf{\Omega_{max}} = \varnothing$, $I = \varnothing$.
FOR $j = 1$ to $|\Omega|$-1 and $j \notin I$
FOR $k = j + 1$ to $|\Omega|$ and $k \notin I$
IF $X_j < X_k$ // X_j is not belonged to $\mathbf{\Omega_{max}}$.
$I = I \cup \{j\}$.
BREAK // Next j
ELSE IF $X_j \geq X_k$ // X_k is not belonged to $\mathbf{\Omega_{max}}$.
$I = I \cup \{k\}$.
END IF
END FOR
$\mathbf{\Omega_{max}} \leftarrow \mathbf{\Omega_{max}} \cup X_j$.
END FOR

Procedure 2
The modified RSDP method to calculate the network reliability in terms of all *D*-MDVs.

FUNCTION RSDP(Ω_{max}).
FOR $i = 1$ to $|\Omega_{max}|$
IF $i = 1$
$R_D = \Pr\{X \le X_i\}$
ELSE
R_D_temp1 $= \Pr\{X \le X_i\}$
END IF
FOR $j = 1$ to i
$X_j = \min(X_i, X_j)$
END FOR
IF $i \ne 2$
X_temp2 = min(X)
R_D_temp2 = RSDP(X_temp2)
END IF
$R_D = R_D + R_D$_temp1 $- R_D$_temp2
END FOR

Fig. 7. A demonstration of the APP development.

R_3, R_4 and R_5, the combination of R_3, R_4 is the best option due to the minimal expected transport time, which is consistent with the results in Tables 9 and 10. Under unclear rescue time situation, the expected transport time can provide another suggestion in an RSS. In practice, the application on smart phone can be developed using the proposed algorithm, which enables the RSS to quickly evaluate the rescue efficiency and further allocate resources in disaster situations. The expected rescue time can be displayed on the user interface (UI) as a tool for the commander. An example of the UI is shown in Fig. 7.

This paper mainly develops an RSS which can be demonstrated by an APP shown as Fig. 7. The stochastic transport time of each rescue road is considered to calculate the probability that casualties can be successfully transported to the CCP, which also represents the network reliability. The proposed algorithm is summarized into three steps. Firstly, the rescue operation is modeled as a network, where transporting casualties from the affected area to the CCP is analyzed. The uncertain damage of rescue road caused by the earthquake can be represented through stochastic transport time, which is calculated with the Pipes' generalized model, and the average transport speed on the damaged road is calculated by the road length, width and the probability of road failure. Next, an algorithm is proposed to find all *D*-MDVs, which are critical to efficiently compute the network reliability. Finally, through experimental results, appropriate decision suggestion such as allocation of the rescue route, number of ambulances, and rescue time plan, can be provided by such performance index. For example, the number of ambulances can be determined by the maximal network reliability which satisfies limited rescue time. The rescue time also can be evaluated by the expected transport time which is calculated by the network reliability. Additionally, sensitivity analysis can be performed to investigate the most critical road that should be maintained first. In particular, the ultimate goal in this paper is to develop an RSS to provide decision-making suggestions. In such safety system, the rescue network can be built, and the algorithm is then proposed to calculate the network reliability, which serves as a reference for evaluating rescue resources and organizing rescue operations to keep the safety of casualties.

This study provides the foundation for evaluating the rescue operation by network reliability. However, further studies are required to accurately represent the uncertain damage dealt to the SRN in earthquake. In the future, researchers can focus on taking more factors into consideration when discussing rescue road uncertainties. Casualties from multiple affected areas can be further considered to expand their practicality. The fragility of bridges and buildings should also be taken into account in cross-county rescue operations.

CRediT authorship contribution statement

Cheng-Hao Huang: Writing – original draft, Validation, Software, Methodology, Data curation. **Yi-Kuei Lin:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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